UNIVERSITY
OF OSLO

Master's thesis

The title of my thesis

Any short subtitle

My Name

Mathematics
60 ECTS study points

Department of Mathematics
Faculty of Mathematics and Natural Sciences

Spring 2025

My Name

The title of my thesis

Any short subtitle

Supervisor:
The Name

Abstract

Here come 3–6 sentences describing your thesis.

This is a relatively short working example of the UiO template for a Master's thesis written in LaTeX. The example is based on Dag Langemyhr's excellent template compliant with the UiO designmanual from 2022, Martin Helsø's instructive examples (2017–2019) and Karoline Moe's seminars on writing in mathematical subjects (2014–2024). We show how some of the standard and most helpful features can be used efficiently, but please note that several of the chapters, sections, packages, tools and environments are optional to include in your thesis and that you must adapt and modify the template to suit your thesis and the traditions in your field of study. If in doubt, ask! Your supervisor and department have the final say in this.

Sammendrag

Her kommer Abstract på et annet språk. Det skal være et sammendrag på norsk i Ph.d.-avhandlinger. Det er like greit å ha det i en masteroppgave også. Det kan være utfordrende å skrive, men det er nyttig.

Dette er en relativt kort tekst der vi bruker UiOs mal for masteroppgaver...

Add new section about results in **??**.

Contents

List of Figures

List of Tables

Preface

The preface is a short and perhaps more personal description of the project, the purpose and your motivation for pursuing it. It should be short and sweet; and it is different from the acknowledgements.

This template has grown out of the work with seminars about scientificness and writing in mathematical subjects that was organized by the Department of Mathematics and the Science Library at the University of Oslo (UiO), and hosted by Karoline Moe, Ph.D., in the period from 2014 to 2024.

There has always been a tradition at the department that former students pass their LaTeX-template on to the next generation of students. However, this practice did not ensure compliance neither with standard traditions in scientific writing and reference management, nor with subsequent design manuals at UiO and typography. Moreover, some of the commands and packages in LaTeX used by students were outdated or in conflict, and there was clearly a need for more instruction.

This template tries to answer these issues. The main purpose of the template is to serve as a starting point for all students using LaTeX to write their Master's thesis. First of all it is a 'minimal' working example with the structure of a thesis. Secondly, We have included examples that most students frequently run into and that can take forever to figure out, even with all available tools, search engines and AI. Finally, we have tried to include a description or short text of the most essential genre-features or pit-falls for each chapter.

We sincerely hope that this template can be of some help in your work, and feel free to send the the LaTeX-group at the University of Oslo Library an e-mail on latexguru@ub.uio.no if you are stuck!

Acknowledgements

chapter:ack

The acknowledgements is where you thank everyone who have contributed to the thesis. It is mandatory to thank your supervisor, co-supervisor, your department and other parties that you have cooperated with; fellow students, a company etc.

The acknowledgements is usually written in a more personal tone than other parts of the thesis, and you should use the pronoun 'I' when you write. F.ex. I would like to thank Dag Langemyhr and Martin Helsø for their passion for LaTeX, their dedication to elegance and detail in typography, and for reading the package documentation at *The Comprehensive TeX Archive Network* (CTAN)[1] so that the rest of us can worry less about layout and more about The Millennium Prize Problems[2].

In the acknowledgements it is custom to include important events that have happened during the period where you were working on the project. Sometimes, this can be done with humor and creativity. But be extremely cautious! Humor changes over time, your life and perspective changes, and you never know who will read this part of your thesis. What seems funny now might be extremely insulting or embarrassing in a couple of years[3].

In fact, the acknowledgements is the part of the thesis that *everyone* who picks up your thesis will read, not just the examiner or the interested researcher. I recommend that you visualize your audience when you feel most creative: your supervisor, your parents, your grandmother, your ex, your future boss, your future co-workers, your future kids, and, perhaps most importantly, your future students.

The fact that your audience might be larger than you think, can not be stressed enough. And this applies to the entire thesis. Your thesis will eventually be deposited in and made openly available to the entire world via the institutional repository DUO. Because of the UiO-address, it gets a high rating on search engines and will be very easy to find for interested researchers all over the world. Moreover, most researchers subscribe to alerts when new citations of their articles or books are published, which means that most of the researchers that you have cited are notified about your thesis, and they might read it, push it to their students, find errors, notice sloppy work and referencing, pick up interesting results and more.

[1] Note that CTAN is the central place for all kinds of material around TeX, with 6649 packages as of October 2024.

[2] Footnotes are usually not used in mathematical texts, in particular not for citations! More on that later. In the meantime, use Oria via ub.uio.no to find books about The Millennium Prize Problems

[3] As a rule of thumb, try to refrain from being too creative and funny in your thesis. Write a blog, diary or send letters to a friend if you must.

Chapter 1

Introduction

intro

1.1 Start with a bang

The introduction is very important in your text because this is where you introduce your research question, put it into context, set the tone of the thesis and usually where a substantial amount of notation and theoretical background is discussed. There will be more of the latter in the background chapter.

Expect to write the introduction several times; early on to get an overview and the last thing you do to ensure that the story in your thesis is coherent.

It is a good idea to start your work by writing the introduction to get more familiar with your topic, the existing research on the topic, and manage citations. Use the sources that you get from your supervisor as a starting point, try to find more literature by reference hunting backwards and forwards in time with a scientific database like MathSciNet or zbMATH, or even Google Scholar. Once you have a decent overview of the relevant research, check if tools like Connected Papers [1] can help you find even more.

Most introductions are written according to the 'and-but-therefore'-principle. Following this recipe - at least for some paragraphs - is an easy way to produce a paragraph/text that is easy to read, engaging and direct to the point.

Once you have finished writing the other chapters, you have to go back and rewrite the introduction to ensure that the introduction actually fits the thesis' structure and introduces the necessary background, notation and correct research question - it tends to change over time.

1.2 Research question

Make sure that you spend time on formulating the research question as precisely as possible. Make it stand out, but avoid descriptive adjectives that are not followed by an argument! Never write 'important, beautiful, interesting, cool, nice etc.' without more information. The sentence 'Here is my very important research question.' is just pointless if you don't explain why.

Question 1. *My main research question is (...).*

[1] Please note that tools like Connected Papers must be used with caution and a critical eye, but sometimes they can be of help.

1.2.1 Citations and reference management

The introduction should point to research leading up to your research question. This means that there will be citations here. Citations and reference management are set up with Biber and the package BibLaTeX, a modern version of BibTeX. There are several options that can be tweaked for this package to make the citations and reference list appear as required in your subfield.

The most used reference styles in mathematical subjects are *alphabetic*, *authoryear* and *numeric*. Alphabetic has the advantage that it produces a unique and permanent identifier for the citation that can be produced with `\cite`. For authoryear, please use `\parencite` or parenthesis around the citation. For numeric, the numbering may be sorted alphabetically or in order of appearance.

The metadata that can indentify an article or book that you cite are coded in a so-called ‘bibitem’, that you can download directly from most databases, preferably standardized (MathSciNet or zbMATH). The bibitems are then collected in the .bib-file. If you’re a pro and use the reference manager Zotero, you can export a .bib-file directly from your library. Please note that the meta-data are not better than the file they are from, so if you use Zotero (or any non-standard database) you must manually quality check and update your bibitems.

To be able to refer back to it, every bibitem must have a cite key, preferably one that you rememeber. Because of the package `Showkeys`, you may freely choose the cite key. However, it is a good idea to save time and working memory - at least during the draft phase of your work - to use the style alphabetic and use the output as cite key.

As in all other fields of science, proper citation and reference management is expected. If a candidate fails to follow the rules, he or she may be accused of plagiarism, cheating or similar, which has huge consequences.

The following is a non-complete list of citation guidelines that apply to mathematics and statistics. This list is based on Kowalski’s excellent online note Citation Guidelines in Mathematics and Krantz’ book [Kra17].

- Give credit to the person behind the idea!
- All your claims must be verified either by a mathematical argument or a citation.
- Citations appear at the beginning of paragraphs, end of sentences, before or in parenthesis next to a result.
- All items in your reference list must have a corresponding citation in the text! There is a surjective function there.
- Due to the need to merge notation or change the level of a result that is cited, a citation is very rarely a direct quotation with quotation marks in mathematics.
- Use primary sources as much as you can.
- It is custom to cite textbooks, and not necessarily primary sources, for well known results.
- Some results are so well known that they do not need a citation at all, but are refered to by name, f.ex. Pythagoras’ theorem.

- If you cite a specific result by someone, add environment and number to the citation: `\cite[Theorem 1.1]{citekey}`. Or add page number(s): `\cite[10]{citekey}` or `\cite[10-20]{citekey}`; this will automatically produce the correct 'p.' or 'pp.' in the citation: [Kra17, Theorem 1.1], [Kra17, p. 10] and [Kra17, pp. 10–20].

- Conversely, if you don't add a citation to a specific result, it is expected that you provide a proof and that it is your own.

- If you build your research on someone's work and only tweak some assumptions, then specify this in the text with a citation and explicitly state the changes that you make.

- It is OK to cite or refer to an entire article as background material, but if you use specific results you must cite these as above. Never ever copy the structure of someone else's text.

- Do not cite friends, supervisors or famous researchers unless you actually use their research; a citation must serve a purpose.

1.3 Outline

The rest of the text is organised as follows. This is an example of a description list; see more lists in **??** and **??** on page 10.

?? consists of an interesting introduction. Zooming in on your research question, taking history into account and clarify what your problem is. You could consider calling this a part if you have multiple result parts in your thesis. It ends with an outline and this is also a good place to place a declaration of AI, but the latter might be placed elsewhere in the thesis.

?? is all about the theoretical background, methods and notation. Don't forget a description of data.

?? is all about your results. It could be theorems and proofs, it could be applications, it could be examples.

?? is more about your results; in subjects that follow the IMRaD-structure, it is known as 'discussion'. Put your results in context. Expand, explain and compare. Weaknesses and strengths of your results. Synergy effects from your results? Perhaps this chapter is not not relevant for you.

?? is your fabulous conclusion. Summing up what you have done. Focus on your contribution and repeat how your results make the world a better place. Point to future reseach.

After the chapters, please print your reference list. This list must include references to all citations in the text, but no more.

New from 2024 is that if you have used AI in any way, this should be decleared in a separate environment, and precise information about the tools used should appear in the reference list.

At the end of the thesis, you include short appendices with code, pseudocode, large tables, more examples or figures. See **??** and **??**. If the length of your appendices is disproportional to the length of your text, consider making it available online with a permanent DOI (Digital Object Identifier), e.g., online repositories like GitHub, Zenodo etc. This will help your future readers and your research group build on your work.

1.4 Statement for the use of AI tools

The use of AI tools should be declared in a statement. There are different suggestions to how this can be done.

The first suggestion below is based on a declaration from the publishing house Elsevier, modified by GPTUiO ([GPT24]).

Declaration of AI-tools I

In the development of this master's thesis, I have utilized the generative artificial intelligence (AI) tools outlined below. I acknowledge full responsibility for all content within this thesis, including sections where AI tools were employed, and for ensuring compliance with the current regulations set by the University of Oslo.

GPTUiO

Name and version GPTUiO, GPT-4 Omni, see [GPT24].

Purpose and extent of use The tool is used to rewrite paragraphs to clarify content in the introduction of this thesis.

Prompt or conversation If you have used AI extensively to generate ideas or text behind your work, which you generally should aviod, please make sure that you add the prompt or conversation in an appendix, see **??**, or at GitHub (or equivalent).

Another suggesion is the following, also based on a declaration from the publishing house Elsevier, and as of 4th February 2025 posted on the UiO web page https://www.mn.uio.no/kurt/english/artificial-intelligence/templates-for-declaration-of-AI-in-assignments.html.

Declaration of AI-tools II

In this scientific work, generative artificial intelligence (AI) has been used. All data and personal information have been processed in accordance with the University of Oslo's regulations, and I, as the author of the document, take full responsibility for its content, claims, and references. An overview of the use of generative AI is provided below.

No AI tools used

If you have not used AI tools, please declare the following:
In the development of this master's thesis, I have not used any AI tools. I acknowledge full responsibility for all content within this thesis and for ensuring compliance with the current regulations set by the University of Oslo.

Chapter 2

Background

background

2.1 Notation

2.1.1 Mathematical typography

Presenting mathematics in a text is quite technical and requires lots of LaTeX code. The first thing you must be aware of is that all mathematics must be placed in math mode to get the correct font. There is a huge difference between n (mathematical symbol) and n (letter). Letters in math mode will always be in math font, so you must specify text: $x = 2 and x = 3$ is a typical error, but either split it in two math modes $x = 2$ and $x = 3$ or add text inside $x = 2 \text{ and } x = 3$.

Please note that mathematics should be included as part of a sentence. It could be inline, like $x = 2$, but then you must avoid line breaks inside the expression. If a mathematical expression is too tall or long for a single line, or it should stand out, it should be displayed centered on a separate line. For instance we have, using the equation environment, that

$$\iint_D \mathrm{d}x\,\mathrm{d}y = \int_0^{2\pi} \int_0^t \rho\,\mathrm{d}\rho\,\mathrm{d}t = \frac{4}{3}\pi^3,$$

which could also be written in display style as

$$\iint_D \mathrm{d}x\,\mathrm{d}y = \int_0^{2\pi} \int_0^t \rho\,\mathrm{d}\rho\,\mathrm{d}t = \frac{4}{3}\pi^3.$$

Notice the differential in ?? typed with the command `\diff`. You can cleverly refer to your equation with `\cref`: ?? or `\vref`: ?? on page eq:integration ?? to get the page number if it is on a different page.

2.1.2 Using symbols

There are two golden rules that applies to the use of symbols in text.

1. Don't introduce unnecessary symbols
2. Define all symbols in the text before they are used

There are three typical mistakes that are made all the time - even experienced researchers do this, and it could quickly escalate into a nightmare if you don't notice it early in the process.

- One thing is referred to by two different symbols

- One symbol means two things
- Breaking the golden rules

When you choose notation, please follow standards in your field.

2.2 Data management plan

Do you have research data? The answer is always ‘yes’, even if you don’t have the typical spread sheet with numbers. Everything you produce, from notes, lab journal, LaTeX, code, figures, images etc. are examples of data. Before you start your project, you should make a plan for all this. Which kinds of data do you expect, how do you plan to store them during and after your project, can they be reused by your research group after you finish etc.?

2.3 Theory

We now state a theorem from [AM69] that has a name and will be used later on. In the code, notice the braces around the precise citation inside the theorem environment.

`thm:dedekind` Theorem 2.3.1 (Dedekind’s theorem [AM69, p. 95]). *Let A be a Noetherian domain of dimension one. Then the following are equivalent:*

1. *A is integrally closed;*
2. *Every primary ideal in A is a prime power;*
3. *Every local ring $A_{\mathfrak{p}}$ $(\mathfrak{p} \neq 0)$ is a discrete valuation ring.*

Notice that the text is in italics in the theorem environment. Sometimes it is convenient to use the command `\noindent` to remove the paragraph indentation immediately after an environment.

Proof. This is what a proof environment looks like. Notice the square □ that marks the end of the proof. You may change this symbol, but stick to standards, e.g., ‘QED’ or ■. □

We could have talked about *Fermat’s last theorem* and cited [Wil95, Theorem 1.1]. Then we want to cite [Vie08], which shows that different values of `sorting` in numeric styles in `biblatex` gives citations numbers in order of appearance or alphabetic.

Chapter 3

Results

results

All chapters must start on a right hand page!

3.1 My first great results involves theorems, lemmas and examples

All results must be given a continuous numbering for cross reference and future citations. This template is set up with a suitable numbering and correct typesetting for the most used theoremstyles in the `uiotools`-file; see this file if you want customize or make a new environment.

Example 3.1.1. This is an example. ▲

Definition 3.1.2. This is a definition.

Theorem 3.1.3. *This is a theorem.*

Proof. This is a proof. □

Remark 3.1.4. This is a remark.

3.2 My second great result involves figures and tables

Figures and tables are important, and they must be included with floats, which makes them pop up wherever LaTeXwants them to. This can seem unfamiliar for a novice LaTeXuser, and requires two things. First, you must include a caption below your figure or above your table containing a short explanation. Secondly, you must comment on your figure or table in your text. This can be done by adding a label underneath the caption in the environment, followed by adding a suitable cross reference in the text.

It is wise to use labels that are easy to recall, preferably systematic with a prefix followed by a description, f.ex. fig:oneball. Using draft mode and the package `showkeys` the labels (and citekeys) are displayed above the text or in the margins, which is very convenient when you work with a large document. This is turned off by adding the option 'final' to the documentclass.

To produce the cross reference, use the command `\cref` (**??**) to call upon the label and an automatic description of the object. If you want a description of the placement of the environment in your cross reference, use `\vref` (**??** on the following page), which will add the location or page number if the figure shows up on a page further away. Read more about cross references in this link [CREF Xio24].

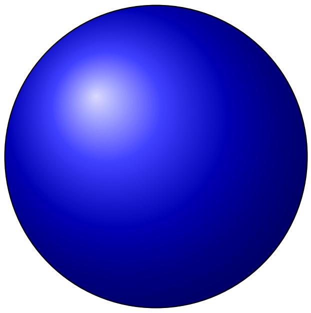

Figure 3.1: This is a figure of one ball, and the caption goes underneath the figure. This could be a complete description of the figure in several sentences. A shorter caption should be made for List of Figures.

fig:oneball

Note that it is possible to tweak the placement of a float using the float specifier option [htbp], where h stands for here, t for top, b for bottom and p for page, but please don't spend time doing that before the week before the due date. Sometimes it is better to include just a little text or move the float in the code.

3.2.1 Figures

Adding a floating environment and presenting a figure, graph or picture can be done in several different ways. The two most common ways are to use `\input`, see **??**, or `\includegrapics`, see **??**.

Note that **??** has a long caption, which is not suitable for the List of Figures at the beginning of the thesis. A shorter caption, more or less a caption-title, should be added as an option.

You can adjust the size of the graphic as you wish by adding options for height and width, crop it and more, but make sure that you have a figure with sufficient resolution, see **??**.

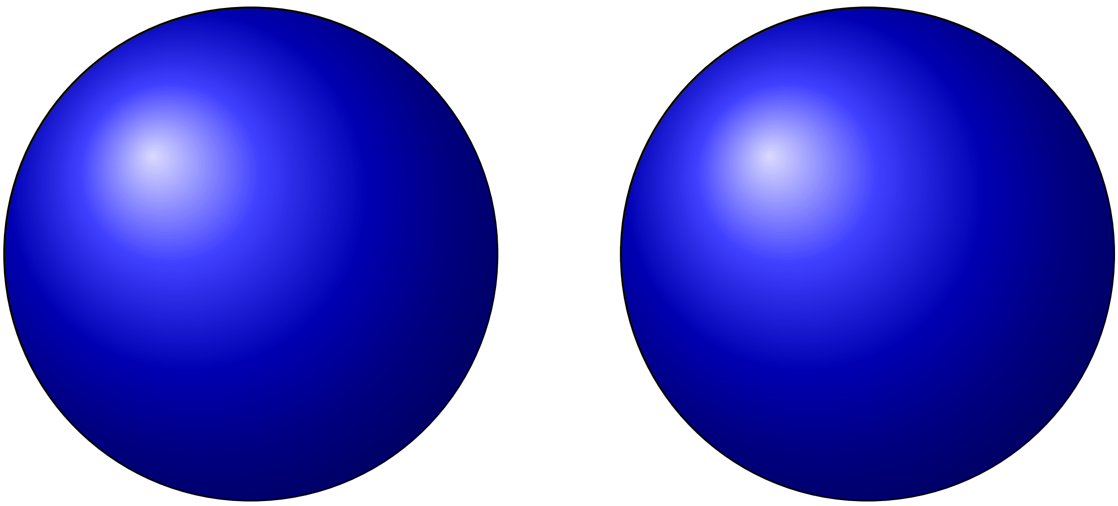

Figure 3.2: Two balls.

fig:twoballs

When you write your thesis, you sometimes just know that you need some text or a figure, but you have not made it yet. Then you can include a todo-note and a missing figure, see **??** on the next page.

I must write a paragraph about this here.

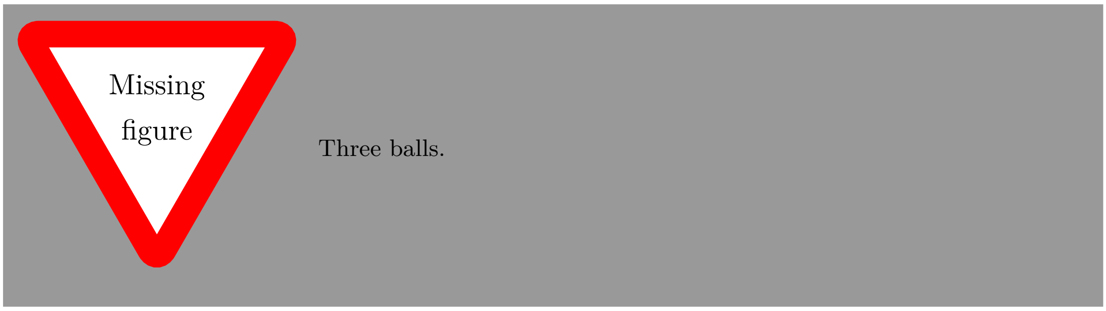

fig:missing

Figure 3.3: I have to make a figure with three balls before I forget how!

tab:colon

Table 3.1: Caption above tables. Proper colon usage. Alternative for List of Tables.

Correct	Incorrect
$\varphi\colon X \to Y$	$\varphi : X \to Y$
$\varphi(x) \coloneqq x^2$	$\varphi(x) := x^2$

Remark 3.2.1. There are so many floats in a row here, that LaTeX struggles to place them, but resist the temptation to tweak the placement before the final touch!

3.2.2 Tables

There is a lot to say about tables, but the main point is that we use the package `booktabs` to create tables with as few lines and good spacing as possible. The first row is usually descriptive and in a different font (bold, textsf etc.). If you need huge or complicated tables with multilines or -columns, try to instruct AI to create the LaTeX-code for this or copy from an online instruction manual; this is not considered plagiarism. Note that there are solutions for tables over multiple pages and tables that extend to the margins, hence need to be rotated 90 degrees.

The examples that follow here, show typical tables with caption above the table - but please notice the important information in the tables as well.

Table 3.2: Proper arrow usage.

Correct	Incorrect
$A \implies B$	$A \Rightarrow B$
$A \impliedby B$	$A \Leftarrow B$
$A \iff B$	$A \Leftrightarrow B$

Table 3.3: Proper quotation mark usage. The \enquote command chooses the correct quotation marks for the specified language. Moreover, \resizebox has been used to control table width.

tab14

Correct	Incorrect
‘This is an “inner quote” inside an outer quote’ (Controlled by ‘babel’.)	’This is an "inner quote" inside an outer quote’

Table 3.4: Proper dash usage. Minus is not minus unless it’s in math mode. Intervals are longer than expected. It is now easy to tell that Birch and Swinnerton-Dyer are two people, but I don’t know why the footnote lands on the wrong page.

Correct	Incorrect
-1	-1
1–10	1-10
Birch–Swinnerton-Dyer conjecture	Birch-Swinnerton-Dyer conjecture
The ball — which is blue — is round. The ball—which is blue—is round.	The ball - which is blue - is round.

Chapter 4

Discussion

discussion

The discussion part is where you discuss your results; implications, strengths and weaknesses.

4.1 All my great results

List with bullets

- This is a list with bullets - the symbol can be changed easily.

! A point to exclaim something!

■ Make the point fair and square.

A blank label?

- This is the last item.

Numbered lists

1. This is a numbered list.
2. This is the second item.

Descriptions

My first great result This is a description list.

My second great result This is the second item.

4.2 Pros and cons

4.3 Future research

Chapter 5

Conclusion

`conc`

A chapter that wraps everything together and sums up the results, concludes from the discussion, points to further research and so on. This might be different in theoretical projects and experimental projects.

5.1 End with a bang!

An easy and structured way to write your conclusion is to do the following exercise:

- Formulate your research question in one sentence.
- Identify the main results from your work and write them down.
- Describe your results in terms of
  - What's new?
  - What's great?
- Describe your results in terms of
  - Weaknesses or disadvantages
  - Future research
- Find synergy and describe significance
  - Combine the three results
  - Implications of your research

References

AM69 [AM69] Atiyah, M. F. and Macdonald, I. G. *Introduction to commutative algebra.* Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 1969, pp. ix+128.

uiogpt [GPT24] GPTUiO. *AI Conversation.* Version GPT-4 Omni. Accessed: 5 November 2024. 5th Nov. 2024. URL: https://gpt.uio.no/.

Kra17 [Kra17] Krantz, S. G. *A primer of mathematical writing.* Second. Being a disquisition on having your ideas recorded, typeset, published, read, and appreciated. American Mathematical Society, Providence, RI, 2017, pp. xx+243. ISBN: 978-1-4704-3658-2. URL: https://doi.org/10.1090/mbk/112.

Vie08 [Vie08] Viefers, S. 'Quantum Hall physics in rotating Bose–Einstein condensates'. In: *Journal of Physics: Condensed Matter* 20.12 (2008), pp. 123–202.

Wil93 [Wil95] Wiles, A. 'Modular elliptic curves and Fermat's last theorem'. In: *Ann. of Math. (2)* 141.3 (1995), pp. 443–551. ISSN: 0003-486X. URL: https://doi.org/10.2307/2118559.

CREF [Xio24] Xiong Ying, L.-T. by. *Advanced LaTeX Cross-references.* Accessed: 10 December 2024. 10th Dec. 2024. URL: https://latex-tutorial.com/advanced-latex-cross-references/.

Appendix A

The First Appendix

The appendix is where you place all information that is not a natural part of your text, but still is relevant to validate or recreate your results. This could be (relatively short) code, large tables, a complete set of examples etc. If you have written a long code in your project, consider using UiO GitHub (Information in Norwegian.) and link it to Zenodo to get a Digital Object Identifier (DOI).

A.1 First Section

We might have to cite some dataset [Aut20].

References

[Aut20] Author, T. *Very Big and Important Dataset*. Dryad, 2020.

Appendix B

Use of AI

B.1 Conversations with UiOGPT

(Consider adding .pdf-files.)

B.1.1 Statement

The following conversation with GPT UiO in [GPT24] has been used in this work:

[User]
Rewrite this text: In the work behind this master's thesis I have used the AI tools as described below. I am aware that I am responsible for all content of this master's thesis, including the parts where AI tools are used, and for ensuring that the thesis complies with current regulations at the University of Oslo.

[GPT]
In the development of this master's thesis, I have utilized the AI tools outlined below. I acknowledge full responsibility for all content within this thesis, including sections where AI tools were employed, and for ensuring compliance with the current regulations set by the University of Oslo.

[User]
Write an equivalent text that indicates that the student has not used AI tools.

[GPT]
In the development of this master's thesis, I have not used any AI tools. I acknowledge full responsibility for all content within this thesis and for ensuring compliance with the current regulations set by the University of Oslo.

References

[GPT24] GPTUiO. *AI Conversation*. Version GPT-4 Omni. Accessed: 5 November 2024. 5th Nov. 2024. URL: https://gpt.uio.no/.